Crack Fractography — where the measurements stand

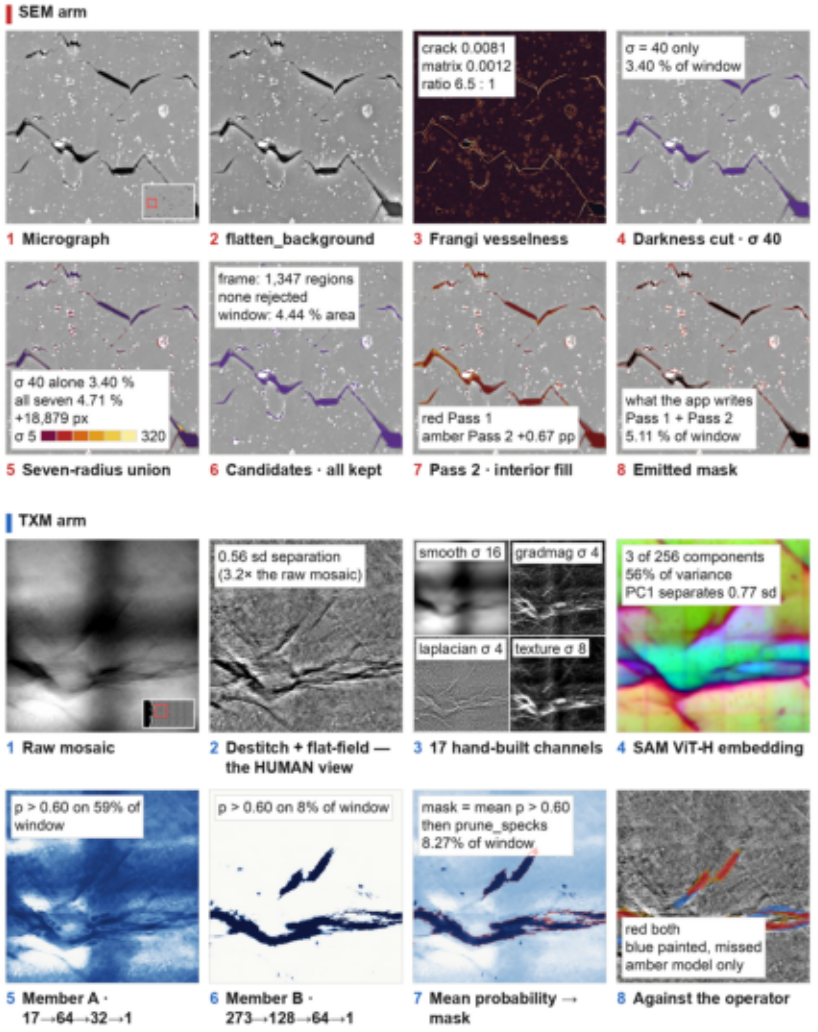
v1.49.2. Every figure here is generated from a committed record by a script in this repository; none is drawn by hand and none carries a number that was not measured.

Caveat. Eight slides. Read each one with its caveat: most of these results were over-read at some point for exactly the reason stated underneath.

What each stage of the detector does

The SEM row is v1.49.2, the pipeline the app runs: seven flattening radii unioned, every candidate region kept.

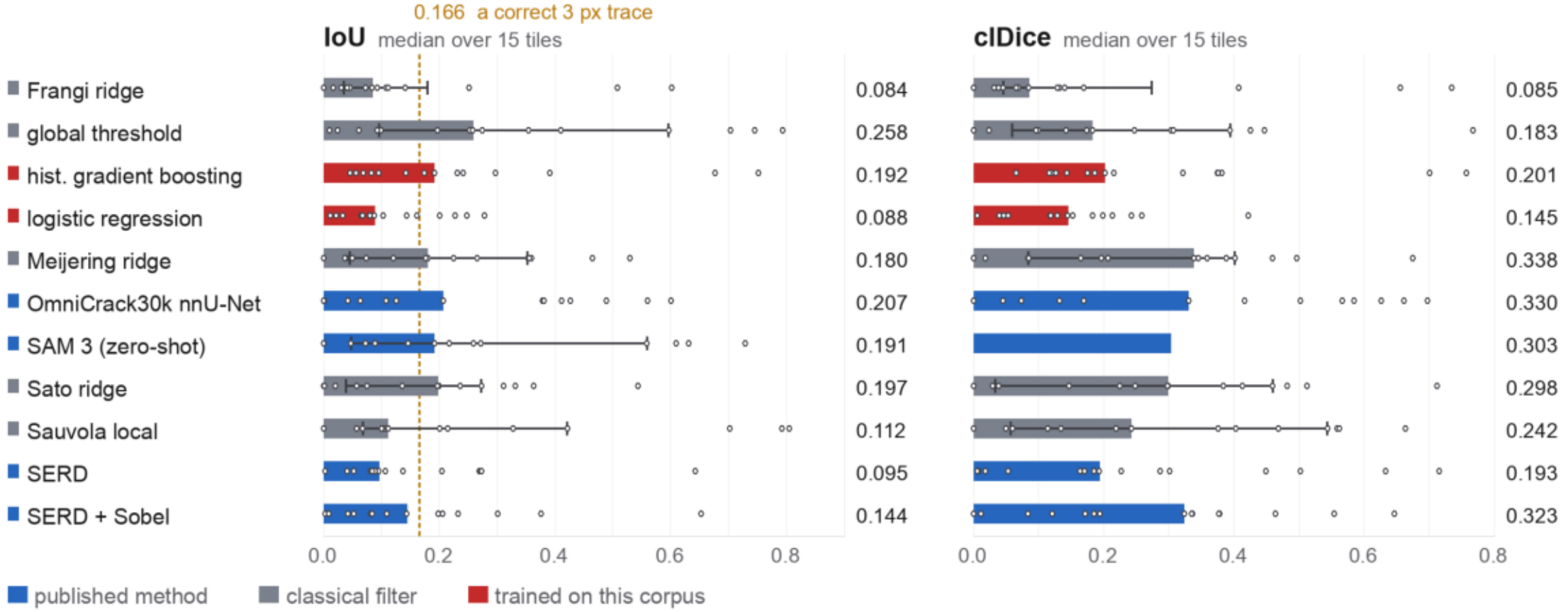
Caveat. Panel 6 is the candidate stage; panel 8 is the full pipeline including Pass 2. The retired region classifier would have removed 21.6% (207,448 px) of what now ships.



Every method on identical data

Eleven arms, the same 15 tiles from the same 9 frames, the same leave-one-frame-out budget. The ranking REVERSES between the two metrics, and that is the result.

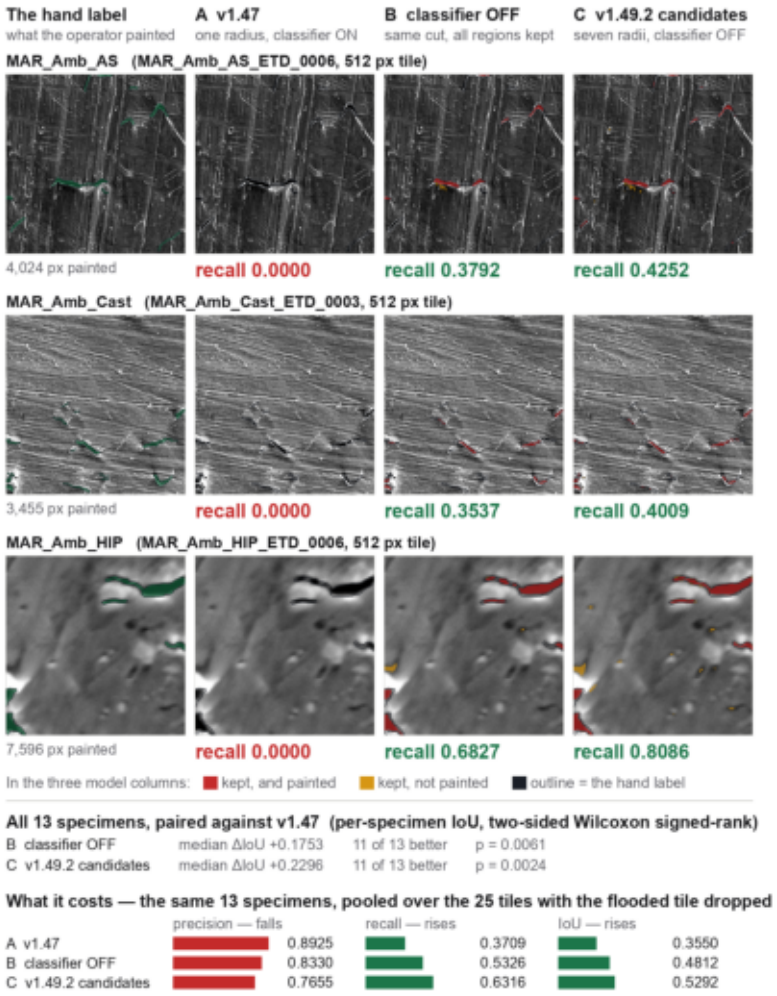
Caveat. The dashed line at IoU 0.166 is what a physically correct 3 px trace scores. Arms above it are not more accurate; they are fatter.



What deleting the region classifier recovers

Three ETD specimens scored recall 0.0000 at every candidate setting ever tried. The candidates covered those cracks all along; the classifier was rejecting them.

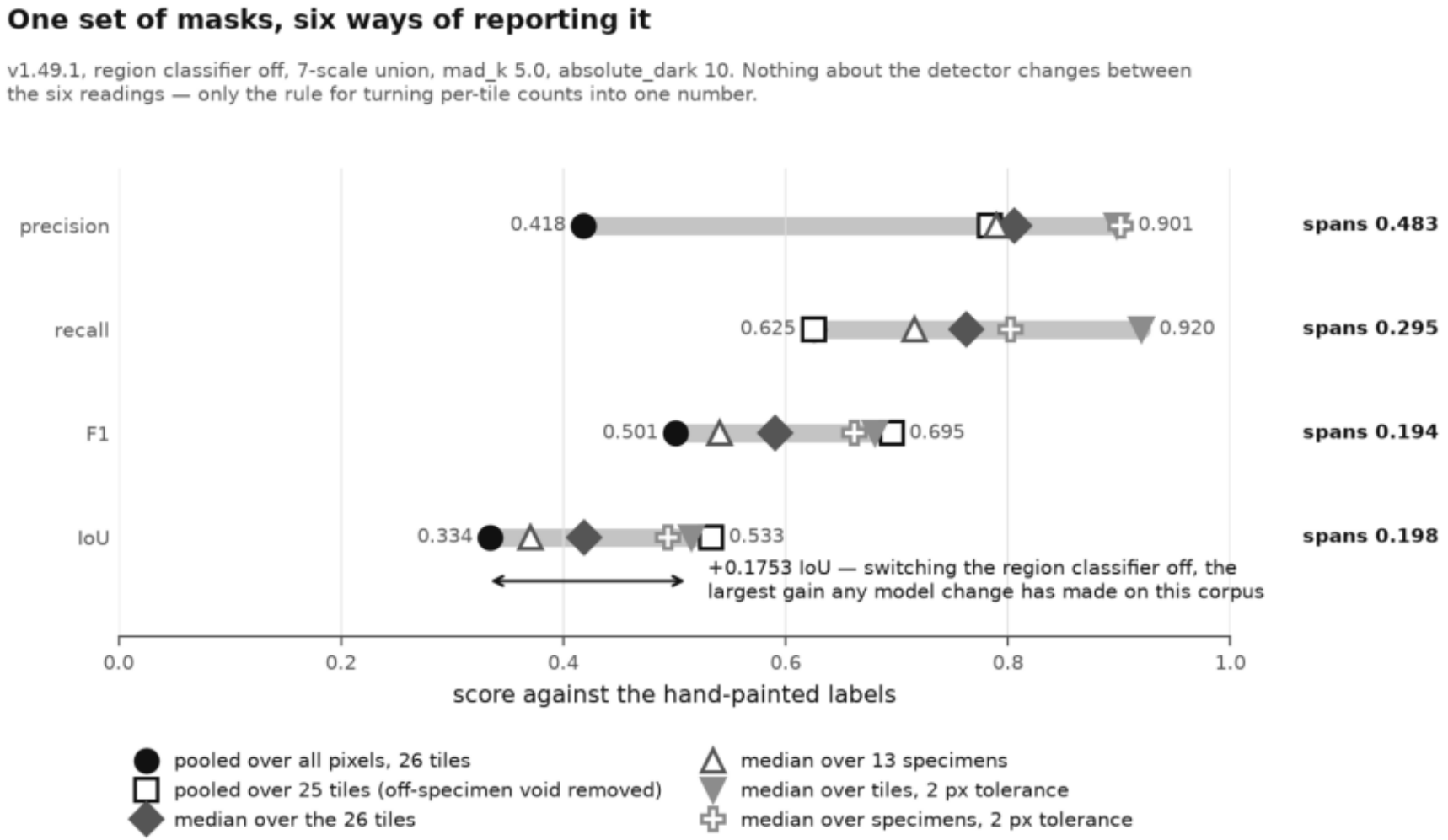
Caveat. It also COSTS precision on ordinary fields, 0.8925 to 0.7840 pooled off the one off-specimen void tile. Shown on the same plate.



The reporting choice moves the number more than any method does

One set of masks, six legitimate ways of reporting it. The IoU span across readings is 0.1985 -- larger than the +0.1753 that switching the classifier off actually bought.

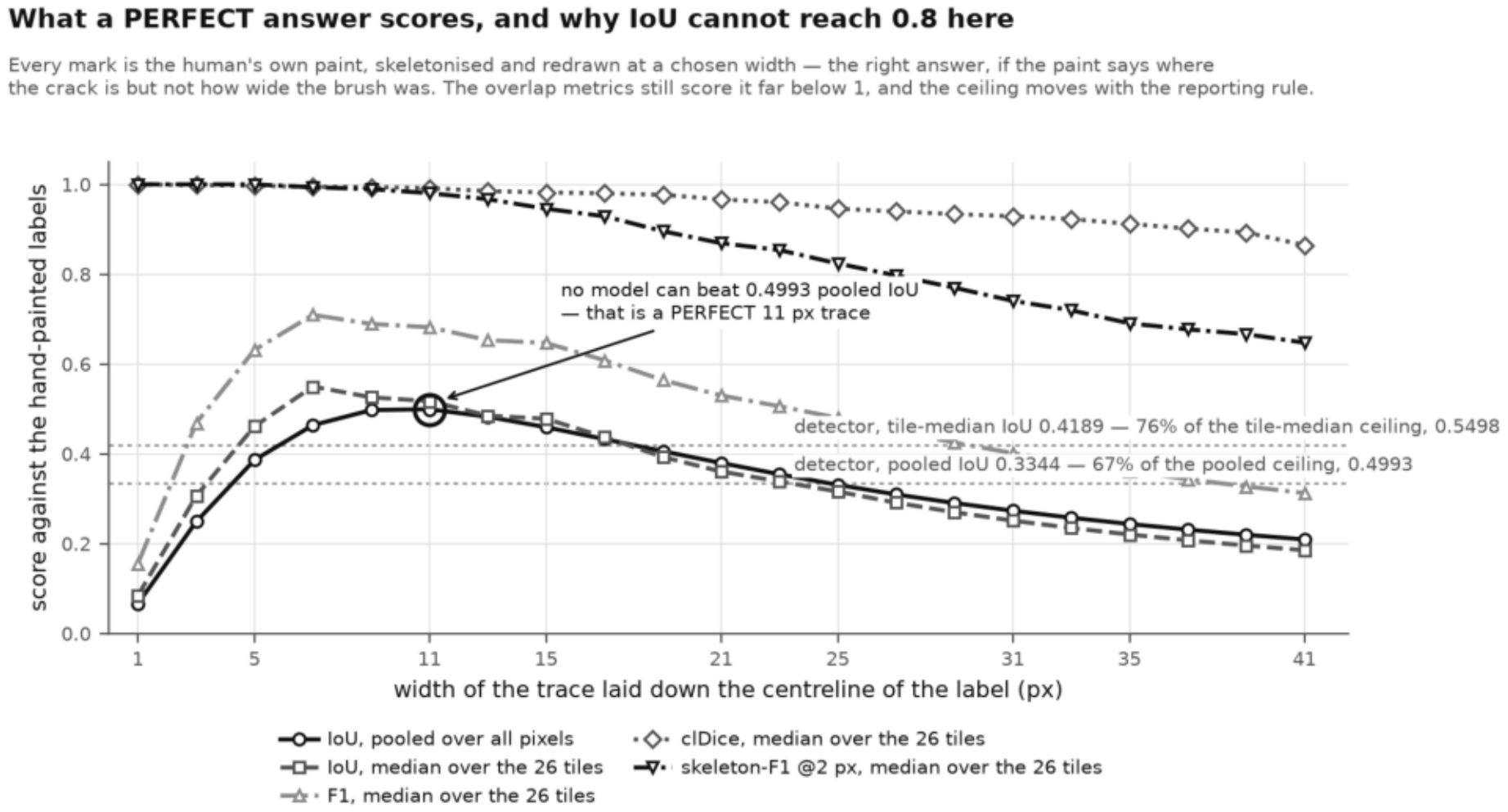
Caveat. Always state the aggregation AND the tolerance. The pooled row and the per-specimen-at-2px row describe the same pixels.



A correct answer caps at IoU 0.50 against these labels

A perfectly placed trace -- the label's own centreline, zero localisation error -- peaks at IoU 0.4993 pooled at an 11 px width, and 0.5498 as a median over tiles at 7 px.

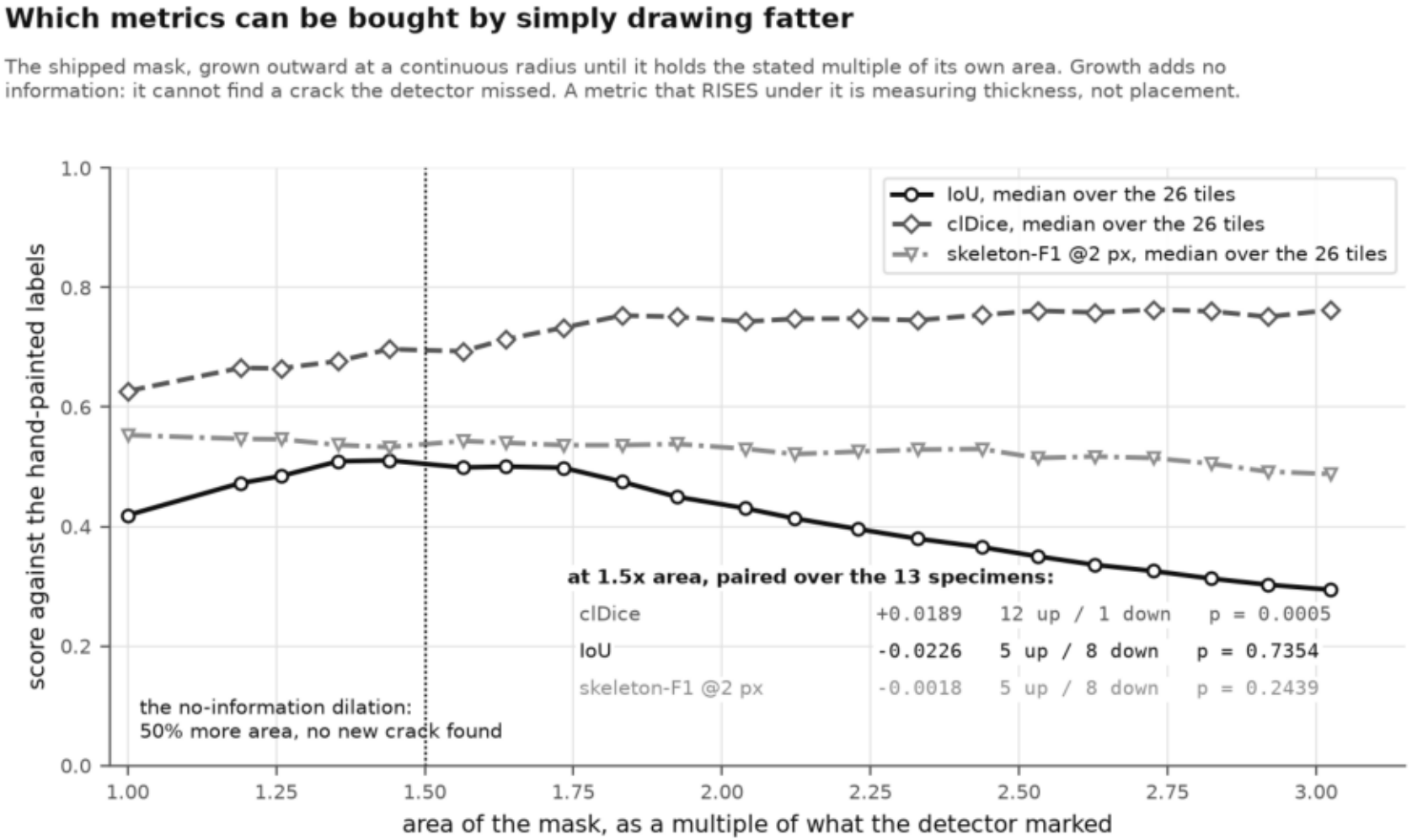
Caveat. So IoU 0.42 is 67-77% of what is achievable, not 42% of perfect. The ceiling is aggregation-dependent too and must be read in the same units as the score.



Both headline metrics reward being fatter than a crack

Dilating the mask to 1.5x its area adds no information and moves clDice +0.0478 on 13 of 13 specimens (p=0.0002). Skeleton-F1 at 2 px moves +0.0015 (p=0.68).

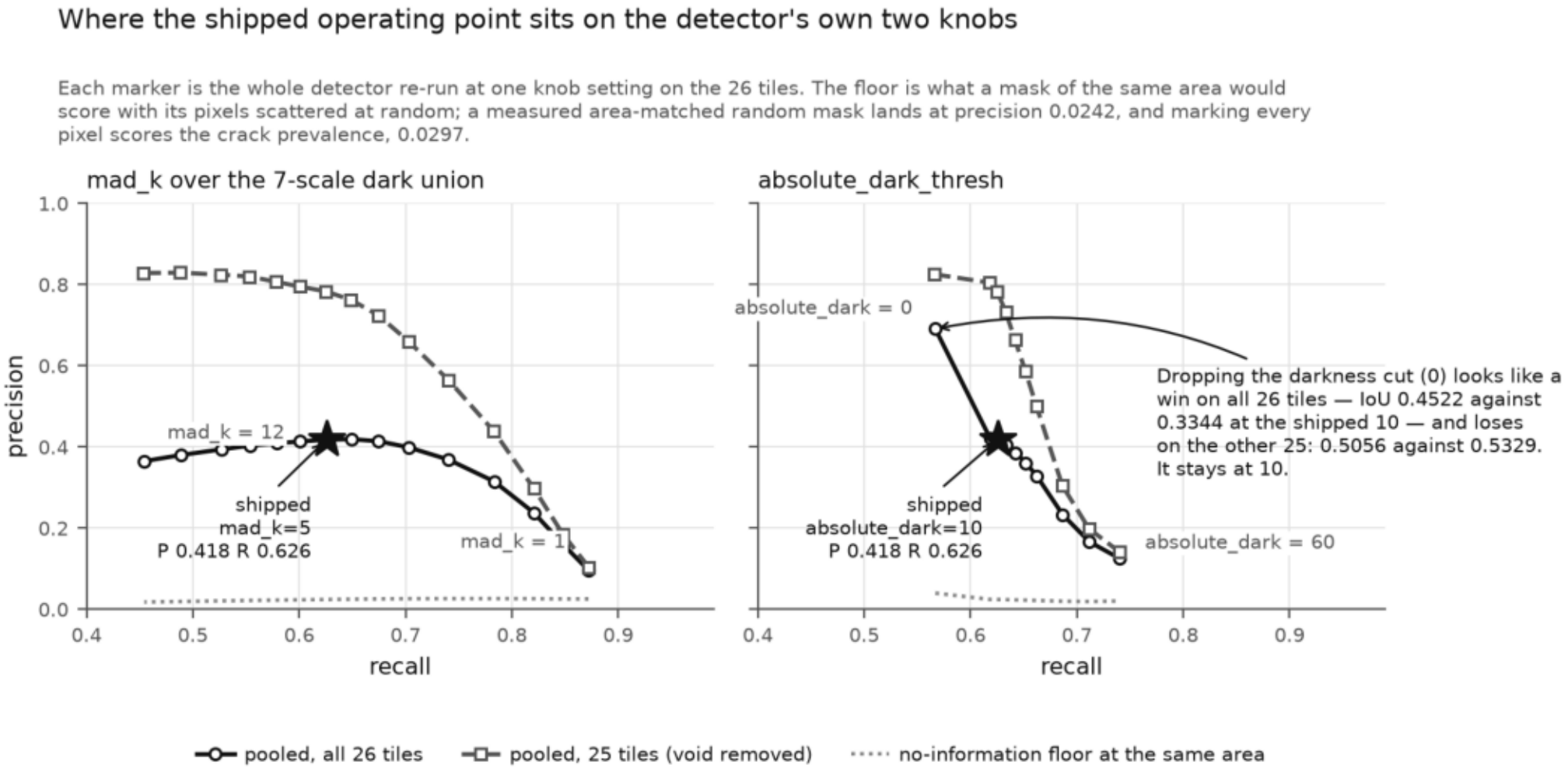
Caveat. The full-frame paint gate has the same disease: a correct 3 px trace of the human's own paint deletes 94.63% of it, because the paint is 154 px wide. Three instruments, one failure mode.



Recall alone is not a score

The loosest setting reaches recall 0.98 and scores F1 0.162. Its precision, 0.0883, is identical to a plain Otsu threshold and is 1.09x a coin flip.

Caveat. Read precision against the no-information floor -- the prevalence of crack among refereed pixels, 0.0810 on TXM and 0.0297 on the SEM tiles.



What each measurement means (1 of 3)

Read straight out of app/groupstats.py, so this slide cannot drift from what the application shows.

Crack count

cracks

How many separate crack regions were found. A COUNT, NOT AN AMOUNT: one long crack and a hundred specks both add to it. Read it beside 'Largest crack's share of area'.

Largest crack's share of area

fraction of crack area

Of all the crack area in a frame, the fraction sitting in the single biggest region. Near 1.0 means one dominant crack; near 0 means a genuinely distributed network. This is what tells you whether the crack count means what you think.

Number density (per megapixel)

cracks / Mpx

Cracks per million pixels of image. Comparable only between frames at the SAME magnification, because a pixel is a different physical size at each.

Number density (per mm2, scale needed)

cracks / mm2

Cracks per square millimetre of specimen surface. The physical version of the row above, so it IS comparable across magnifications -- but only frames with a recovered pixel size contribute.

What each measurement means (2 of 3)

Read straight out of app/groupstats.py, so this slide cannot drift from what the application shows.

Total crack area (pixels)

px2

Every crack pixel in the group added up. Summing this across frames of different magnification adds areas that are not the same physical size.

Total crack area (um2, scale needed)

um2

The same total converted to physical area. Only frames with a known pixel size contribute, so this can cover fewer frames than the pixel row above it.

Crack area fraction

% of frame

What percentage of the imaged area is crack. Dimensionless, so it is the one size measure that compares fairly across magnifications and modalities.

Average crack area

px2

Mean area of one crack region. Pulled upward by a single large crack; compare it with the median below.

What each measurement means (3 of 3)

Read straight out of app/groupstats.py, so this slide cannot drift from what the application shows.

Median crack area

px2

The typical crack's area -- half are smaller, half larger. Unlike the mean, one huge crack cannot move it.

Average length (all)

px

Mean crack length along its centreline, INCLUDING cracks that run out of the frame. Those are measured only as far as the frame edge, so this is a lower bound.

Average length (uncensored only)

px

The same average over only those cracks fully inside the frame, so no length is cut short. If this is much larger than the row above, the frame is clipping your cracks.

Average aspect ratio

length / width

How elongated a region is: length divided by width. Around 1 is round -- porosity rather than a crack. Painted crack in this corpus runs about 3.3; painted porosity about 1.5.

How good is this model, against published ones?

On the 15-tile benchmark the published models were scored on, the shipped detector leads both metrics. The lead survives dropping the frames whose tiles informed its configuration, and does not survive dropping the specimens - n=2 there, which settles nothing either way.

Caveat. Every competing arm is scored leave-one-FRAME-out, which penalises it for having a tunable parameter. v1.49.2 has no parameter fitted on these tiles, but 3 of the 9 frames and 5 of 7 specimens also appear in the 26-tile set used to choose its configuration, so the all-15 row is the favourable reading. And at n=15 nothing on this benchmark separated statistically in the first place.

arm	IoU	clDice	basis
>>> shipped v1.49.2, all 15 ti	**0.3628**	**0.4246**	**no parameter fitted here; config infor**
>>> shipped v1.49.2, frame-o	**0.3530**	**0.3658**	**n=9; frames never used to choose the**
>>> shipped v1.49.2, specim	**0.1596**	**0.2883**	**n=2; too small to conclude, trends ag**
global threshold	0.2579	0.1827	one line of code, LOFO
NESTED LOFO ensemble	0.2526	0.1739	honest nested selection
Sato ridge	0.1972	0.2981	classical filter, LOFO
SAM 3 (zero-shot union)	0.1914	0.3030	foundation model, no fitting
OmniCrack30k nnU-Net	0.2069	0.3304	trained on 30,017 images incl. steel
Meijering ridge	0.1802	0.3382	classical filter, LOFO; best clDice of the cor
Sauvola local	0.1117	0.2421	classical filter, LOFO
Frangi ridge	0.0843	0.0849	classical filter, LOFO
a correct 3 px trace	0.1662	0.9997	the physically correct answer; IoU ceiling r